

AI INTERVIEW PREP SERIES DAY 1: AI BASICS

AI • ML • Deep Learning • Learning Types • AI Lifecycle

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AGENDA

- AI, ML & Deep Learning Fundamentals
- Types of Machine Learning
- Real-World AI Applications
- Training vs Inference
- AI Lifecycle
- Interview Tips & Key Takeaways

WHAT IS AI?

The simulation of human intelligence by machines.

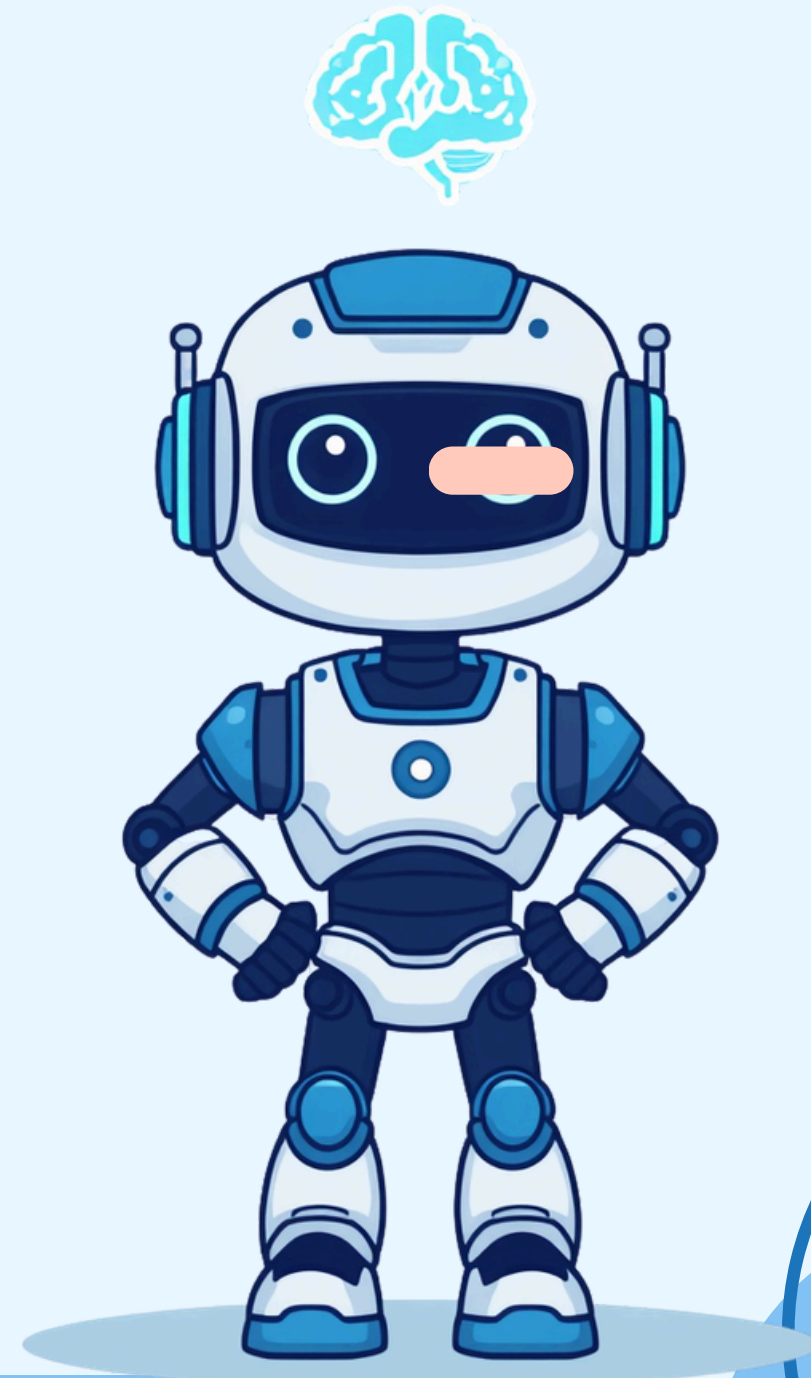
Key Capabilities:

- ◆ Learning
- ◆ Reasoning
- ◆ Decision Making
- ◆ Problem Solving

Examples:

- ChatGPT
- Self-Driving Cars
- Virtual Assistants
- Recommendation systems

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WHAT IS MACHINE LEARNING?

A subset of AI that learns from data.

Key Characteristics:

- ◆ Learns Patterns
- ◆ Makes Predictions
- ◆ Improves Over Time

Examples:

- Spam Detection
- Fraud Detection
- Recommendation Systems



WHAT IS DEEP LEARNING?

A subset of Machine Learning that uses Artificial Neural Networks.

Key Characteristics:

- ◆ Learns Complex Patterns
- ◆ Multiple Hidden Layers
- ◆ Requires Large Datasets
- ◆ High Computational Power

Applications:

- Image Recognition
- Speech Recognition
- NLP & LLMs



AI VS ML VS DL

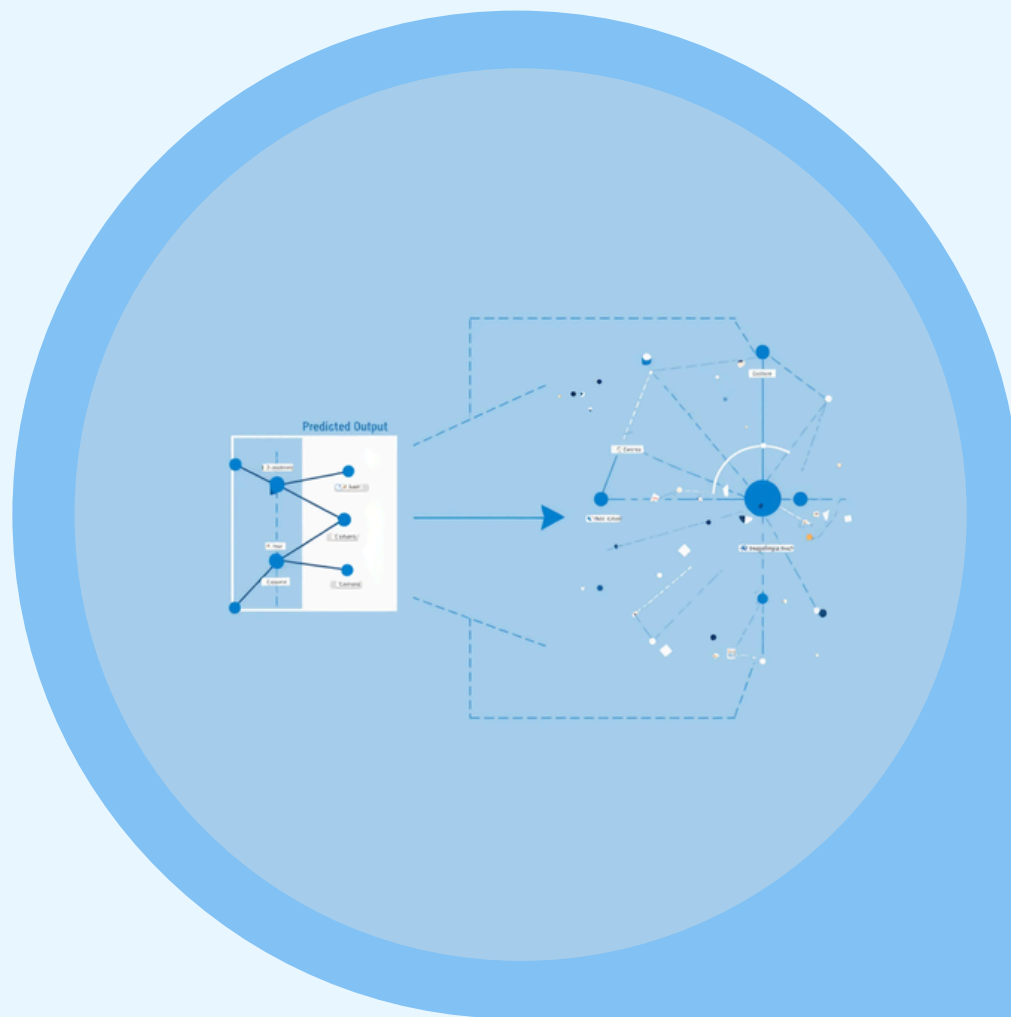
- **AI (Artificial Intelligence)** : Broadest concept — machines simulating human intelligence.
- **ML (Machine Learning)** : Subset of AI — systems that learn from data.
- **DL (Deep Learning)** : Subset of ML — uses neural networks with many layers.



TYPES OF LEARNING

- ◀ Supervised Learning – Learns from labeled data.
- ◀ Unsupervised Learning – Finds patterns in unlabeled data.
- ◀ Reinforcement Learning – Learns through rewards and penalties.

SUPERVISED LEARNING



Uses Labeled Data

- Learns from input-output pairs
- Predicts outcomes for new data

Key Examples:

- Spam Detection
- House Price Prediction
- Image Classification

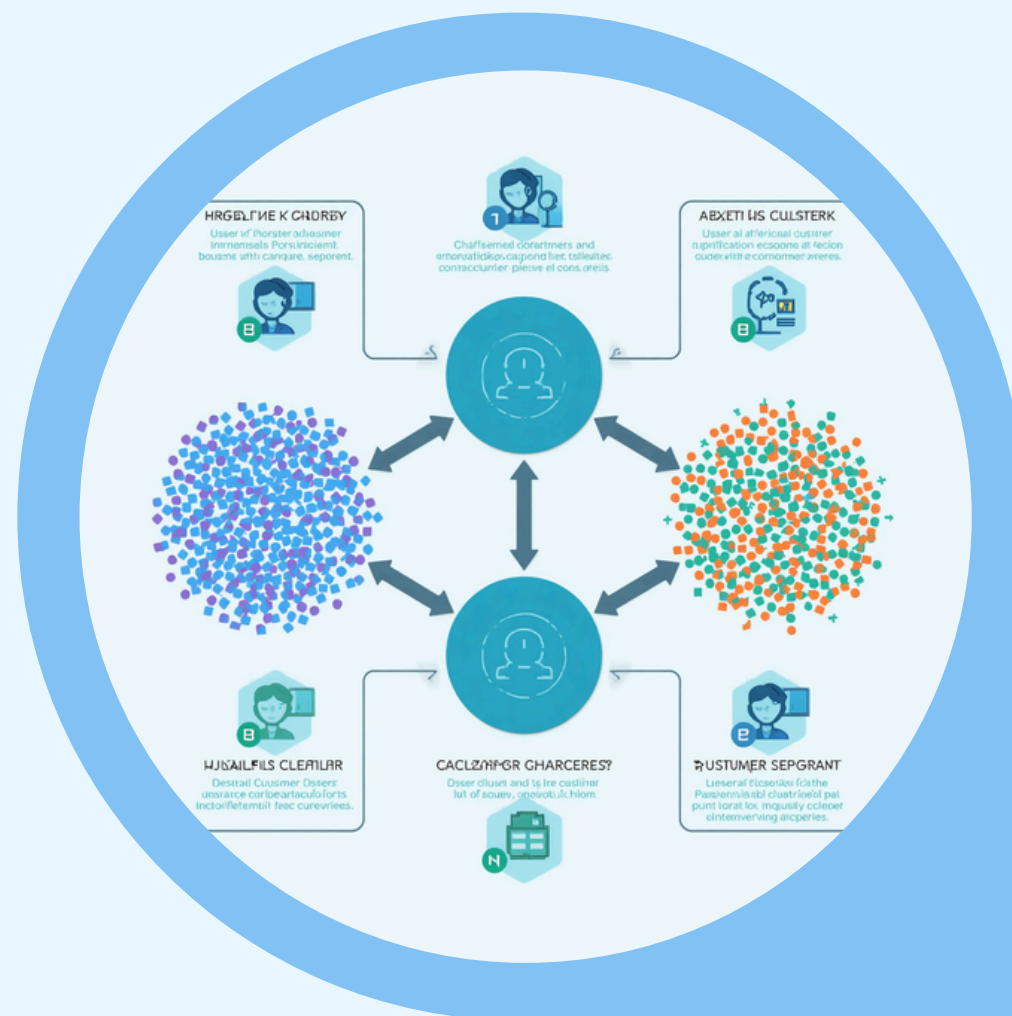
UNSUPERVISED LEARNING

Uses Unlabeled Data

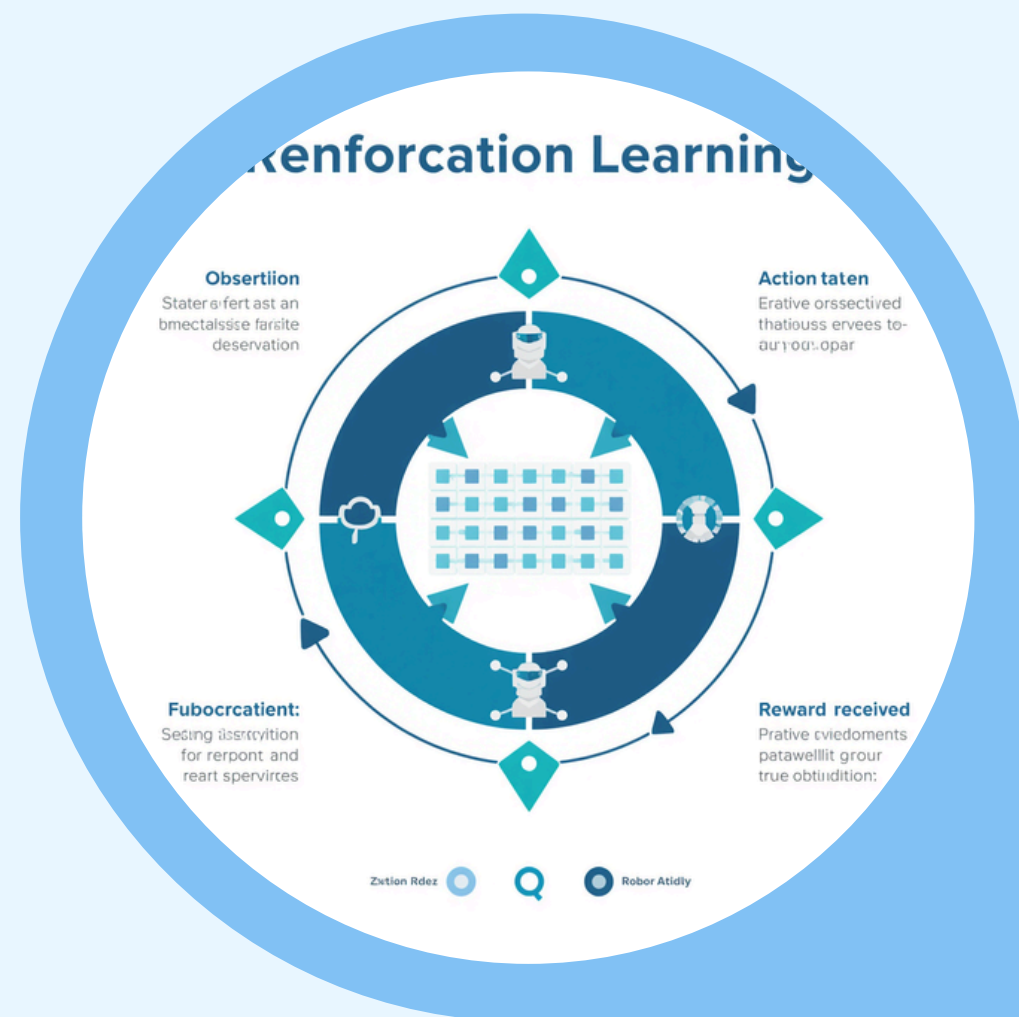
- No predefined answers
- Discovers hidden patterns and relationships
- Identifies clusters, groups, or anomalies

Real-World Examples:

- Customer Segmentation
- Anomaly Detection
- Topic Modeling
- Recommendation Systems



REINFORCEMENT LEARNING



Learns Through Rewards and Penalties

- Agent interacts with an environment
- Receives rewards for correct actions
- Receives penalties for incorrect actions
- Learns the best strategy over time

Examples:

- Game AI (Chess, AlphaGo)
- Robotics
- Autonomous Vehicles
- Navigation Systems

REAL-WORLD AI APPLICATIONS



Healthcare

- Disease Diagnosis
- Medical Imaging Analysis



Computer Vision

- Self-Driving Cars
- Object & Lane Detection



Recommendation Systems

- Netflix
- Spotify
- YouTube



Natural Language Processing (NLP)

- ChatGPT
- Google Translate
- Virtual Assistants

TRAINING VS INFERENCE

Training

- Model learns from data
- Patterns are identified
- Requires large datasets and compute

Example:

Teaching a model to recognize cats using thousands of images.

Inference

- Model makes predictions on new data
- No learning takes place
- Fast and efficient

Example:

Uploading a new image and the model predicts whether it contains a cat.



AI LIFECYCLE

1. Data Collection
2. Data Preparation
3. Model Training
4. Evaluation
5. Deployment
6. Inference
7. Monitoring

Key Idea:

AI systems continuously learn, improve, and evolve over time.

KEY TAKEAWAYS

- ✓ AI is the broad field of intelligent systems
 - ✓ Machine Learning is a subset of AI
 - ✓ Deep Learning is a subset of Machine Learning
 - ✓ Three main learning types:
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
 - ✓ AI systems follow a lifecycle:
Data Collection → Training → Deployment → Monitoring
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NEXT UP: DAY 2

Machine Learning Fundamentals – See You on Day 2!

